CIVITAS indicators

Parcel locker throughput (TRA_FR_LK2)

DOMAIN

Transport	Environment	Energy	Society	Economy

TOPIC	**Freight**	
IMPACT	**Parcel lockers for urban deliveries** *Increased use of shared parcel locker infrastructure to consolidate last-mile drop-offs and collections and reduce the number of separate delivery trips.*	**TRA_FR**

Category

Key indicator	**Supplementary indicator**	State indicator

CONTEXT AND RELEVANCE

Parcel lockers consolidate last-mile drop-offs and collections at a shared, unattended location, allowing a single vehicle visit to serve multiple recipients instead of separate door-to-door deliveries. This can reduce the number of delivery stops, failed delivery attempts, and re-delivery trips associated with individual home deliveries. The extent to which these benefits materialise depends not only on the number of lockers installed, but on the volume of parcels handled relative to the compartment capacity available over time: a network with low throughput delivers little consolidation benefit, while one with higher throughput can make more intensive use of shared infrastructure.

This indicator measures parcel locker throughput: the number of parcels deposited per available locker compartment per day, within a defined geographical area and reference period. **It is a relevant indicator when the policy action aims to increase the volume of parcels handled through parcel lockers. A successful action is reflected in a <u>HIGHER</u> value of the indicator after the experiment compared to the BAU case.**

DESCRIPTION

The indicator assesses the average number of parcels deposited in parcel lockers for each operational compartment available per day. The indicator should be calculated for the parcel lockers in the pilot area and over a defined reference period. Each parcel is counted once when deposited, i.e., its collection is not counted as a second transaction.

The indicator is expressed as **parcels per compartment per day**.

METHOD OF CALCULATION AND INPUTS

The indicator should be calculated **exogenously** based on the required inputs and then coded in the supporting tool**.**

Method 1	
Survey parcel locker operators/providers	Significance: **0.25**

INPUTS

The following information is needed to compute the indicator:

- The **total available compartment-days** across all parcel lockers in the pilot area during the observation period, calculated by summing the number of operational compartments available on each day. If capacity is constant, this equals the number of compartments multiplied by the number of observation days.
- The **total number of parcels deposited** in the lockers during the observation period. Each parcel should be counted once at deposit, regardless of when it is collected.

METHOD OF CALCULATION

The indicator should be computed **exogenously** according to the following steps:

- **Collect the number of parcels deposited and available compartment-days from parcel locker operators/providers, then divide the former by the latter.**

EQUATIONS

The equation that should be applied to quantify the indicator is:

$$\boldsymbol{ThPL = \frac{ParcelsPL}{CompDaysPL}}$$

Where:

$ThPL$ = Parcel-locker throughput (parcels per compartment per day)

$ParcelsPL$ = Total number of parcels deposited in the lockers during the observation period

$CompDaysPL$ = Total available compartment-days across the parcel lockers during the observation period

ALTERNATIVE INDICATORS

This indicator measures parcel lockers throughput. To assess instead the share of deliveries made via lockers, the CIVITAS Evaluation Framework provides indicator **TRA_FR_LK1**: Share of deliveries to parcel lockers and drop-off points.